

5	(a)	(i)	The <u>centripetal force is provided by the gravitational force</u> acting on stars A and B. By <u>Newton's Third Law</u>, the <u>gravitational force acting on stars A and B</u> are of <u>equal magnitude and opposite direction</u>. Hence, the centripetal force acting on both stars has the same magnitude.	1 1
		(ii)	$\omega = \frac{2\pi}{T}$ $= \frac{2\pi}{4.0 \times 365 \times 24 \times 60 \times 60}$ $\omega = 5.0 \times 10^{-8} \text{ or } 4.98 \times 10^{-8} \text{ rad s}^{-1}$	1 1
	(b)	(i)	Since the centripetal force acting on star A and B are of equal magnitude. $M_A \omega^2 d = M_B \omega^2 (2.8 \times 10^8 - d)$ $\frac{M_A}{M_B} = \frac{(2.8 \times 10^8) - d}{d} = 3.0$ $4d = 2.8 \times 10^8$ $d = 7.0 \times 10^7 \text{ km}$	1 1 1
		(ii)	Gravitational force provides for the centripetal force acting on star A. Hence, $\frac{GM_A M_B}{(2.8 \times 10^8 + 10^3)^2} = M_A \omega^2 d$ $M_B = \frac{(4.98 \times 10^{-8})^2 (7.0 \times 10^7 \times 10^3) (2.8 \times 10^{11})^2}{6.67 \times 10^{-11}}$ $M_B = 2.0 \times 10^{29} \text{ kg}$	1 1 1
6	(a)	(i)	Thrust and Weight	1